Dynamic Disclosure with(out) Timestamps*

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Abstract

We study the role of timestamps in a dynamic disclosure game with an evolving state. At a random date, a sender privately obtains one piece of hard evidence of a hidden and evolving binary state. The sender's flow payoff is the receiver's posterior belief that the state is good. When evidence carries a timestamp, in the unique equilibrium, the sender discloses good evidence immediately and bad evidence after a deterministic, timestamp-dependent delay. Without timestamps, we first show that good evidence is not disclosed immediately unless the prior belief is sufficiently low; moreover, under some conditions, an equilibrium exists in which bad evidence is never disclosed. We then construct an equilibrium, unique within its class, that features up to three phases: a stockpiling phase in which the sender delays disclosure, followed by a purging phase with stochastic disclosure of evidence at a timestamp-independent rate, and then a phase of immediate disclosure of any good evidence. Timestamps prevent the sender from pretending good evidence is fresh and allow the sender to prove that bad evidence is old, thereby accelerating good-evidence disclosure and facilitating bad-evidence disclosure.

1 Introduction

A key feature of many evidentiary documents is whether they carry a timestamp. Evidence may therefore convey not only information about the underlying state, but also information about when that information was generated. Medical test results, financial filings such as bankruptcies and audit reports, and legal records like signed contracts and digital forensic logs all bear verifiable dates. Yet many commonly used documents are not timestamped. For

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example, undated recommendation letters provide an evaluation without indicating when it was written; sensationalist news media sometimes reuse old photos or videos to support current narratives when original dates are difficult or costly for the public to verify.

There has been growing attention in the literature to how the timing of disclosure affects beliefs and incentives in dynamic environments. However, existing work typically abstracts from whether the act of disclosure itself reveals the time at which the evidence was generated. It remains an open question whether the presence or absence of timestamps plays a role in strategic disclosure over time.

We address this question by introducing and studying a simple continuous-time sender-receiver model of dynamic disclosure of hard evidence about an evolving state. There is a hidden state θ_t that evolves according to Markov switching dynamics between two states, 1 ("good") and 0 ("bad"). A sender privately obtains one piece of hard evidence at some random, exponentially distributed date τ and can choose to disclose it (or not) at any later time. We consider two distinct environments: *timestamps* and *no timestamps.* In the timestamps environment, upon disclosure, the receiver sees both θ_τ and τ; in the no-timestamps environment, the receiver only sees θ_τ and must infer τ itself. The sender's flow payoff is the receiver's posterior belief that the state is 1.

This simple environment generates rich equilibrium dynamics that differ qualitatively between the timestamps and no-timestamps environments. To illustrate the effect of timestamps on strategic disclosure of good evidence, first suppose that the bad state is absorbing. In this case, disclosing bad evidence leads to a continuation payoff of 0 and is strictly dominated. In the timestamps environment, immediate disclosure of good evidence is a dominant strategy. If the sender obtains good evidence $\theta_\tau = 1$ and discloses it immediately, the receiver's belief jumps to 1 and decays according to the hidden Markov switching dynamics thereafter. If instead the sender were to delay until some time $t > \tau$, the receiver would retroactively update about θ_τ, so his belief would be the same at all times $s \geq t$, but it would be lower during the interval $[\tau, t)$.

By contrast, in the no-timestamps environment, we show that for some parameter values equilibrium must feature delay. Specifically, continue to suppose the bad state is absorbing, and suppose the prior belief about the state is high. Toward a contradiction, suppose the receiver conjectured immediate disclosure of good evidence at all times, and the sender happens to obtain good evidence at time 0. By disclosing it immediately, the receiver's belief makes a small upward jump (to 1) and begins decaying due to the Markov switching. By instead disclosing after a small delay, the receiver's belief is lower in the interim, but, importantly, it still jumps to 1 after disclosure since the receiver does not observe the timestamps and assumes the evidence is fresh, and then it begins decaying. Thus, delay involves a trade-off between foregoing a higher belief in the short run and inducing a slightly higher belief at all times in the long run. When the prior belief is high, this trade-off favors delay.[1]

For high prior beliefs, we construct an equilibrium consisting of three distinct phases, which we label *stockpiling*, *purging*, and *transparency*. In the stockpiling phase, the sender does not disclose good evidence, creating a stockpile of (possible arrival dates of) undisclosed good evidence from the receiver's perspective. At some date, the sender would prefer to disclose good evidence. However, equilibrium cannot have a bang-bang disclosure strategy with respect to good evidence, with an abrupt transition to a phase of immediate disclosure, because at the first instant of disclosure, the receiver would be expecting all old good evidence to be disclosed, and therefore would not revise his belief to 1. The sender could simply wait an instant longer and separate from these older evidence types. This tension — where a conjecture too much disclosure incentivizes delay — is naturally resolved with mixing in the purging phase. During this phase, the sender stochastically discloses old evidence at an intensity that depends on calendar time but not on the evidence's timestamp. Over time, the disclosure intensity increases which causes the stockpile to deplete, shifting the distribution of evidence to more recent dates. This means that disclosure is eventually interpreted more favorably, which sustains the sender's indifference to delay in the first place. By the end of this phase, the disclosure intensity explodes and the stockpile fully (but continuously) empties. At that point, the receiver's belief is sufficiently low that the sender is willing to disclose any good evidence immediately.

These results have clear policy implications. Without timestamps, there can be delayed disclosure of good evidence, so institutions that require timestamps can help the receiver get information sooner.

The sender's incentive to delay in the purging phase depends not only on the receiver's belief about the current state, but also on the receiver's beliefs about the sender's evidence, since the latter belief determines the receiver's belief after disclosure and the evolution of the receiver's belief conditional on no disclosure. Technically, this presents a challenge in that the receiver's current belief about the state is not the only state variable. Moreover, the receiver must form a belief distribution at each instant about the continuum of past evidence arrival dates. However, with timestamp-independent disclosure rates, we show that these beliefs can be summarized by a few stock variables that track the joint probability that the sender has good or bad evidence and the state is good, conditional on no disclosure. We

[1] In standard disclosure models, equilibria often feature cutoff strategies, with highest types disclosing immediately. The delayed disclosure in our model does not contradict this theme, because the notion of "type" applies not only to whether evidence is good or bad but also its timestamp, and from this perspective, there is no highest type — since for any arrival time of good evidence, there are always future arrival times for the sender to mimic.

describe the rest of the construction in detail in Section 4.1.

When the bad state is *not* absorbing, our model also creates scope for strategic disclosure of bad evidence. The mechanism is that, since evidence only arrives once, once bad evidence has been disclosed, the receiver no longer expects good evidence to be disclosed. This shuts down a "no news is bad news" effect which can offset the direct negative effect of bad evidence disclosure.[2] Again, the presence of timestamps has important implications for the nature of bad evidence. To illustrate, suppose the good state is absorbing; this isolates the problem of bad evidence disclosure because immediate disclosure of good evidence becomes a dominant strategy. In the timestamps environment, we rule out the existence of an equilibrium in which the sender always withholds bad evidence, and show that the unique equilibrium strategy is to disclose bad evidence after a deterministic, timestamp-dependent delay.[3] The key feature of timestamps is that they allow the sender to prove that old bad evidence is in fact old. Therefore, when the sender discloses old bad evidence, beliefs update retroactively downward to 0, but drift up according to Markov switching after that and without the "no news is bad news" effect. For each bad evidence arrival date, there is a date at which the receiver's belief after disclosure of that evidence is exactly the same as it was just prior to disclosure, and the post-disclosure belief path is strictly greater. This is the optimal time to disclose bad evidence. Since the sender discloses at the first instant it is myopically optimal to do so, the equilibrium is independent of the sender's discount rate.

Finally, we analyze disclosure of bad evidence without timestamps, and we find that the role of timestamps is more subtle. We provide sufficient conditions such that there is an equilibrium in which bad evidence is never disclosed, supported by a worst-case off-path belief threat that such evidence is fresh. Hence, in some cases, the policy implication is reinforced: timestamps can facilitate the disclosure of bad evidence.[4]

Related literature

To our knowledge, our paper is the first to study the role of timestamps in a dynamic disclosure setting.

Our paper contributes to a large literature on strategic disclosure of hard information by a sender without commitment, pioneered by Grossman and Hart (1980), Grossman (1981), and

[2]We conjecture that this mechanism would also be present, albeit dampened, in a model that allows evidence to arrive more than once.

[3]In contrast to the no-timestamps environment, the timestamp is payoff relevant.

[4]Our sufficient condition holds for a larger range of parameter values when the sender is very patient. Intuitively, when the sender is very impatient, she places a large weight on the immediate negative effect of bad evidence disclosure, and is less incentivized to disclose. In contrast, impatience increases the sender's incentive to disclose good evidence early.

Milgrom (1981). While the canonical static models predict full unraveling, with essentially all sender types disclosing to separate themselves from lower types, Dye (1985) showed that when there is uncertainty about whether the sender has evidence to disclose, unraveling need not be complete, as low types with evidence can refrain from disclosing it to mimic high types that do not have evidence. A long literature on mostly static models has explored various alternative explanations for nondisclosure.

A small and growing recent literature has studied disclosure in dynamic settings. In Acharya, DeMarzo and Kremer (2011), a sender chooses when to disclose information about a fixed state of the world when there is exogenous public news about that state. The authors show that exogenous news creates a real options problem for the sender: by waiting to disclose, the sender retains the option to not disclose in case very favorable news arrives.

In Guttman, Kremer and Skrzypacz (2014), the sender can obtain multiple pieces of evidence sequentially across two stages, but the state (firm value) is constant and independent of the evidence arrival times. The authors show that late disclosures are interpreted more favorably. Our focus is different; evidence arrival time and the state are intimately linked due to Markov switching. Specifically, more recent good evidence is more indicative of the state being good currently, and the opposite is true of bad evidence. This allows us to study optimal disclosure timing even when there is only one piece of evidence.

Kremer, Schreiber and Skrzypacz (2024) study disclosure of evidence about an evolving state. There, the sender must either disclose evidence immediately or never; late disclosure is not permitted, which is consistent with Securities and Exchange Commission rules. In our model, delayed disclosure is permitted and arises in equilibrium for both good and bad evidence.

Other papers study dynamic persuasion with commitment; e.g., Ely (2017) and Ely, Georgiadis and Rayo (2025). In the latter paper, feedback is given to the sender (receiver) according to a bang-bang rule: first silence, then transparency. Although the results are not directly comparable due to fundamental model differences, it is notable that in our no-timestamps environment, the bang-bang policy cannot occur in equilibrium, and instead there is a purging phase between our stockpiling (silence) and transparency phases.

2 Model

The game is played between a sender and a receiver in continuous time over an infinite horizon. There is a hidden, binary underlying state $\theta_t \in \{0, 1\}$ that evolves according to Markov switching; when the state is i, it switches to $j \neq i$ at constant exponential rate λ_i. Neither the sender nor the receiver observes the realization of the state process, and they share a common prior over the initial state $p_0 = \Pr(\theta_0 = 1)$.

At some random date τ, the sender privately obtains hard evidence of the current state θ_τ. The receiver does not directly observe τ or θ_τ. We assume that τ is distributed exponentially with parameter μ and is independent of the state process $(\theta_t)_{t\geq 0}$. The sender can receive evidence at most once. The sender then chooses to disclose this evidence at any date $s \geq \tau$ or never disclose it, which we denote by $s = +\infty$. We consider two environments: *timestamps* and *no timestamps*. In the timestamps environment, when evidence is disclosed, the receiver observes both the realization θ_τ and the timestamp τ. In the no-timestamps environment, the receiver only observes the realization θ_τ (and must infer τ).

A strategy for the sender consists of a collection of cumulative distribution functions $H(\cdot;\theta_\tau,\tau,t)$ on $[t,\infty]$, for $\theta_\tau \in \{0,1\}$ and $0 \leq \tau \leq t$, where $H(\cdot;\theta_\tau,\tau,t)$ specifies the probability that a sender who obtained evidence θ_τ at time τ and has not disclosed it before time t plans to disclose by time s.

A belief process for the receiver is a $[0,1]$-valued càdlàg process $p = (p_t)_{t\geq 0}$ adapted to the public filtration, where p_t denotes the receiver's posterior belief at time t that $\theta_t = 1$.

The sender obtains a flow payoff equal to p_t and discounts the future at constant rate $r > 0$.

A perfect Bayesian equilibrium (henceforth, equilibrium) is a strategy and belief process p such that the sender's strategy is sequentially rational and the belief process is consistent with Bayes' rule.[5]

It is useful to define $\phi_t := p^* + (p_0 - p^*)e^{-(\lambda_0+\lambda_1)t}$ as the ex ante probability that $\theta_t = 1$ based on the Markov switching dynamics, where $p^* := \frac{\lambda_0}{\lambda_0+\lambda_1}$ is the long-run steady state belief.

3 Timestamps

3.1 Equilibrium Characterization

There exists a unique equilibrium. In this equilibrium, if the sender receives good evidence $\theta_s = 1$ at time s, the sender discloses it immediately at time s; if the sender receives bad evidence $\theta_s = 0$ at time s, the sender delays disclosure until time $s + a^*(s)$, where $a^*(s) \in [0,\infty)$. In particular, the optimal disclosure time, $s + a^*(s)$, is the date the receiver's belief upon observing evidence timestamped at s coincides with the belief upon no disclosure, but leads to a strictly higher belief afterward.

[5]Since evidence arrives stochastically, nondisclosure is always on path, and Bayes' rule applies. Disclosure can be off-path. In the timestamps case, the receiver's belief after off-path disclosure is already determined by the timestamp. In the no-timestamps case, there is flexibility in the off-path belief.

Proposition 1. *There exists a unique equilibrium. If the sender ever possesses good evidence (on or off path), he discloses it immediately. If he possesses bad evidence from time s, he discloses it at time $s+a^*(s)$, where $a^*(s) \in [0,\infty)$ for all $s \geq 0$ is the unique positive solution to*

$$\frac{\lambda_0+\lambda_1}{\lambda_0} p_0 e^{-\mu(s+a^*(s))} = \int_s^{s+a^*(s)} \mu e^{-\mu\tau} \left(e^{-(\lambda_0+\lambda_1)\tau} \left(p_0 - \frac{\lambda_0}{\lambda_0+\lambda_1} \right) - \frac{\lambda_1}{\lambda_0+\lambda_1} \right) \left(1 - e^{-(\lambda_0+\lambda_1)(s-\tau)}\right) d\tau. \quad (1)$$

If the sender still possesses this evidence after time $s + a^*(s)$ (off path), he discloses it immediately.

In words, given

The analysis of equilibrium in the timestamps environment depends on two key features. First, regardless of when it is disclosed, on or off the equilibrium path, the receiver will update his belief (retrospectively) based on the timestamp. Moreover, because the sender can only get one piece of evidence, once he discloses it at some time $\hat{t}$, the receiver does not expect more disclosure afterward and will update her belief post-disclosure according to the Markov process, $\Pr(\theta_t = 1|\theta_s = i)$ for $i = 0,1$ and all $t \geq \hat{t}$.

The figure below illustrates the equilibrium belief dynamics under a high prior (left panel) and a low prior (right panel). Suppose the sender receives bad evidence $\theta_{s_1} = 0$ at time s_1. In equilibrium, the sender optimally discloses this evidence at time $s_1+a^*(s_1)$. The receiver's belief moves along the black curve up to s_1 and then follows the red curve for $t \geq s_1$. If the sender receives good evidence at s_1, she discloses it immediately, causing the receiver's belief to jump to 1 and subsequently follow the blue curve for $t \geq s_1$. The belief upon no disclosure converges to a value strictly below the long-run steady state belief $p^* = \lambda_0/(\lambda_0+\lambda_1)$. This is because there is always a positive probability that the sender got bad evidence recently and is optimally delaying its disclosure.

To see the intuition behind the equilibrium, consider the following best response argument. For disclosing good evidence, fix the sender's strategy of disclosing bad evidence with a delay. If the sender discloses evidence immediately, the receiver's belief jumps up to 1 immediately and decreases according to the Markov switching afterward, where "no news is no news." Delaying disclosure does not improve the post-disclosure belief: upon disclosure, the timestamp leads the receiver to update retrospectively to the exact same belief path. However, delaying strictly lowers the receiver's belief prior to disclosure. Because bad evidence is disclosed with a delay, the absence of disclosure always carries a "no news is bad news" component. This means the belief upon no disclosure is lower than the belief after

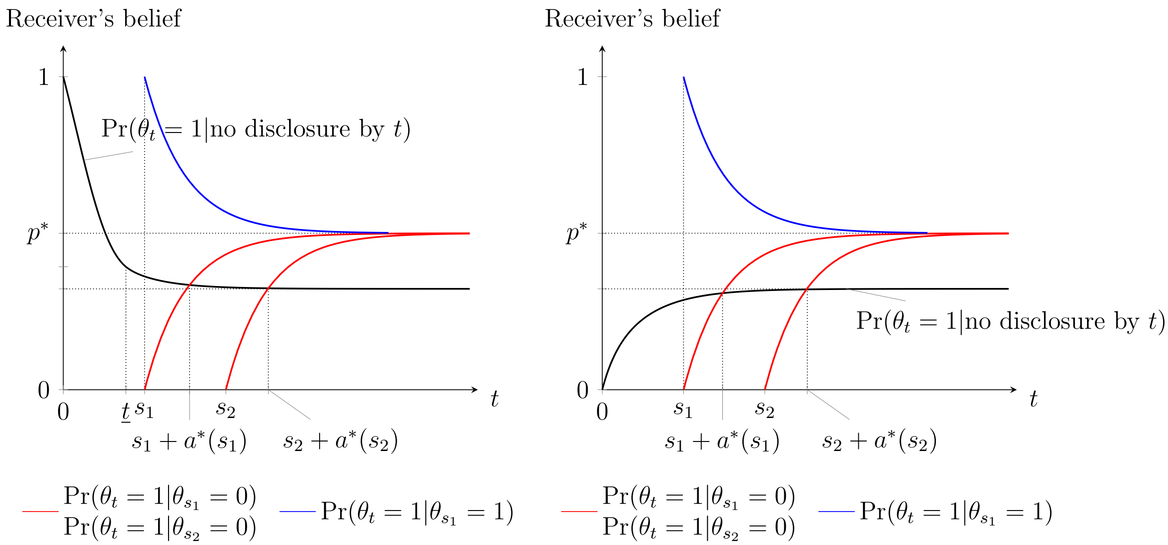

Figure 1: Equilibrium belief evolution for priors $p_0 = 1$ (left panel) and $p_0 = 0$ (right panel), and parameters $\lambda_0 = 1, \lambda_1 = 1, \mu = 0.5$, and $r = 0.1$.

disclosing good evidence, where "no news is no news."

For good evidence, the key feature of timestamps is that they prevent the sender from pretending old evidence is fresh: if good evidence is disclosed with a delay, the receiver's belief updates to the same belief as if it were disclosed immediately.

The logic for delaying disclosure of bad evidence is more subtle. Fix the sender's strategy of disclosing good evidence immediately. Under this strategy, never disclosing bad evidence is not optimal. To see why, suppose toward a contradiction that bad evidence is never disclosed. Over time, absence of disclosure becomes increasingly uninformative of the current state: the probability that some evidence has arrived increases, so no disclosure is more likely to come from the sender withholding bad evidence than not receiving good evidence. Consequently, the belief upon no disclosure converges to the ex ante long-run steady state belief but remains below the "no news is no news" belief due to the persistent (albeit small) probability that the sender has not received good evidence.

By contrast, when bad evidence is disclosed, the receiver's belief drops initially but then drifts upward over time due to Markov switching. Since evidence arrives at most once, disclosure eliminates the possibility of future good evidence and thereby shuts down the "no news is bad news" effect. This means the post-disclosure belief eventually exceeds the no-disclosure belief, making it profitable for the sender to deviate to disclosing at the point of intersection.[6]

For bad evidence, the key feature of timestamps is that they allow the sender to credibly

[6]This argument is formalized and illustrated in Claim **??** in the Appendix.

show that bad evidence is old, making disclosure optimal after some time when its implication for the current state is not as pessimistic.

3.2 Equilibrium Properties

Figure 2 below illustrates the optimal disclosure delay $a^*(s)$ as a function of the time s at which the sender receives bad evidence. Although $a^*(s)$ is monotone in s, the direction of monotonicity depends on the prior. When the prior is high, earlier arrival of bad evidence implies a longer delay. The receiver's belief upon no disclosure declines over time but remains high initially. So early in time, it takes longer for it to fall to the same level as the belief upon disclosing bad evidence. By contrast, when the prior is low, the receiver's belief upon no disclosure trends upward. In this case, earlier arrival of bad evidence leads to a shorter optimal delay. This is summarized in the following observation.

Observation 1. *For all $s \geq 0$, (i) $a^*(s)$ is monotone in s, (ii) $\lim_{s\to\infty} a^*(s) = a^*$, and (iii) $s + a^*(s)$ is monotone in s.*

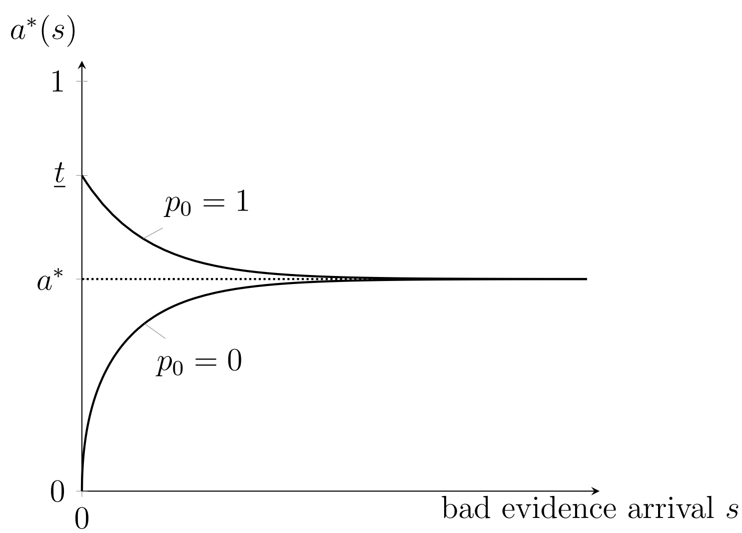

Figure 2: Optimal delay $a^*(s)$ as a function of s for priors $p_0 = 1$ and $p_0 = 0$, and parameters $\lambda_0 = 1, \lambda_1 = 1, \mu = 0.5$, and $r = 0.1$.

3.3 Special Cases: Absorbing States

The equilibrium characterization crucially depends on the presence of timestamps. To study the role of timestamps, we turn to the no-timestamps environment in the next section. To isolate the effect of timestamps on the two types of evidence, we consider two extreme parameter cases: $\lambda_0 = 0$ when focusing on good evidence, and $\lambda_1 = 0$ when focusing on bad evidence. We characterize the equilibrium in these special cases in the timestamped environment, which will serve as benchmarks for comparison with the no-timestamps setting.

The equilibrium characterized in Proposition 1 continues to apply in these special cases, with appropriate simplifications. If $\lambda_0 = 0$, the bad state is absorbing. It is therefore a dominant strategy to never disclose bad evidence: disclosure drives the receiver's belief to 0 permanently, while no disclosure generates a strictly positive belief. Given that bad evidence is never disclosed, it remains optimal to disclose good evidence immediately. If $\lambda_1 = 0$, the good state is absorbing. It is therefore a dominant strategy to always disclose good evidence immediately: disclosure makes the receiver's belief jump up to 1 permanently, while no disclosure generates a belief strictly below 1. The strategy to disclose bad evidence with delay $a^*(s)$ remains optimal and is the unique optimal strategy.[7]

4 No Timestamps

Without timestamps, good evidence need not be disclosed immediately in equilibrium. In a conjectured equilibrium with immediate disclosure of good evidence at all times, the sender can have an incentive to delay disclosure to pretend the evidence is fresher. We begin with a simple, but general, result that says that immediate disclosure occurs only if the prior belief is sufficiently low; for higher priors, there *must* be delayed disclosure of good evidence in equilibrium, in contrast to the timestamps environment.

Proposition 2. *If $p_0 > \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}$, then there is no equilibrium with immediate disclosure of good evidence at all times.*

To understand Proposition 2 and the cutoff belief, suppose the sender obtains good evidence at time 0. By disclosing this evidence immediately, the posterior belief jumps to 1 and then decays toward the long run steady state based on the Markov switching dynamics. If the sender deviates and discloses at some small time $\epsilon > 0$, then on $[0, \epsilon)$, the posterior belief is lower than had the sender disclosed at time 0. Since beliefs are right-continuous, to a first-order approximation, the loss from this delay is $(1 - p_0)\epsilon$. However, when the receiver sees the disclosure at time ϵ, she updates to 1 assuming that the evidence is fresh; the belief then proceed to decay in the same way as before. In other words, delay results in a lower belief for the receiver in the short run, but a higher belief at all times after the disclosure. Compared to having disclosed at time 0, the belief at time ϵ is higher by approximately $\lambda_1\epsilon = 1 - (1 - \lambda_1\epsilon)$, where $-\lambda_1$ is the slope of the post-disclosure belief at time

[7] Uniqueness is shown in Lemma **??** in the Appendix.

0. Moreover, this higher belief results in a higher belief at all future times, albeit discounted, and the net present value of this difference is $\lambda_1 \epsilon \frac{1}{\lambda_0+\lambda_1+r}$. Put together, delay is optimal if $(1-p_0)\epsilon < \lambda_1 \epsilon \frac{1}{\lambda_0+\lambda_1+r}$, which rearranges to the condition in Proposition 2.

Although Proposition 2 implies that equilibrium must feature delayed disclosure of good evidence, it does not say *how* such delay occurs. To that end, and to isolate the role of good evidence, suppose $\lambda_0 = 0$; that is, the bad state is absorbing. (We return to $\lambda_0 > 0$ later in the section.) Then disclosure of bad evidence is strictly dominated (with or without timestamps), since it guarantees a flow payoff of 0 forever after.

The following class of equilibria is central to our analysis.

Definition 1. *A three-phase equilibrium is characterized by two deterministic times* $\underline{T}, \overline{T}$ *with* $0 \leq \underline{T} \leq \overline{T} < \infty$ *such that*

- Stockpiling phase: *On* $[0, \underline{T})$*, there is no disclosure of good evidence. Belief updating immediately after off-path disclosure of good evidence is passive.*
- Purging phase: *On* $(\underline{T}, \overline{T})$*, the sender discloses good evidence at rate* $\kappa_t > 0$ *independent of its timestamp, where* κ *is continuous.*
- Transparency phase: *On* $[\overline{T}, \infty)$*, the sender discloses good evidence immediately.*

We now present the main result of this section, which establishes a unique equilibrium within the three-phase class. Define $\overline{p} := \frac{r+\frac{\lambda_1}{2}}{r+\lambda_1}$ and $\underline{p} := \frac{r}{r+\lambda_1}$.

Proposition 3. *Assume* $\lambda_0 = 0$*. For all* p_0, λ_1, μ *and* r*, there exists a unique three-phase equilibrium. Moreover,*

- *If* $p_0 > \overline{p}$*, all three phases are nonempty.*
- *If* $p_0 \in (\underline{p}, \overline{p}]$*, then only the purging and transparency phases are nonempty.*
- *If* $p_0 \leq \underline{p}$*, then only the transparency phase is nonempty.*

Figure 3 illustrates the equilibrium beliefs when $p_0 > \overline{p}$; p_t (with some abuse of notation) denotes the receiver's belief about θ_t conditional on no disclosure, and q_t denotes the receiver's posterior immediately after disclosure of good evidence.[8] In the stockpiling phase, delay leads to an accumulation or stock of undisclosed evidence from the receiver's perspective. The receiver's belief p conditional on no disclosure falls due to Markov switching, but beyond

[8]Note that while the belief jumps to q_t at time t, after that, it evolves according to Markov switching; i.e., the receiver's belief does not continue to be given by q at all future times after the disclosure.

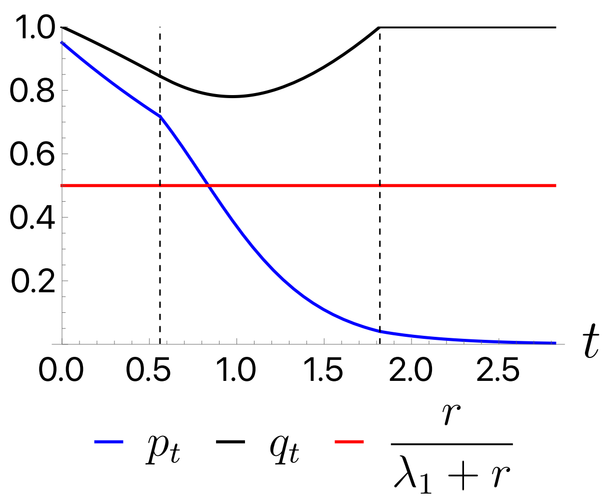

Figure 3: Belief paths in the three-phase for $p_0 > \overline{p}$, with vertical dashed lines at $\underline{T}$ and $\overline{T}$.

that, "no news is no news" as the receiver anticipates delay. Meanwhile, q falls due to the aging of evidence, but this fall is dampened by the arrival of new evidence.

The purging phase begins at the time when, if the receiver had counterfactually continued to conjecture delay, the sender would strictly prefer to disclose. However, delay followed by a strict preference to disclose at time $\underline{T}$ cannot be sustained in equilibrium. If the receiver conjectured such a bang-bang disclosure strategy, then after disclosure at time $\underline{T}$, the receiver's belief would jump to a value less than 1, based on its belief distribution of past arrival dates of good evidence. By waiting until time $\underline{T} + \epsilon$ to disclose, for $\epsilon > 0$ arbitrarily small, the sender can induce a posterior belief arbitrarily close to 1. This marginal delay yields a discrete gain at arbitrarily small cost, so the sender would prefer to deviate. Mixing in the purging phase resolves this dilemma.

At the beginning of the purging phase, p has a downward kink as the rate of disclosure is bounded away from 0. During this phase, q initially continues to fall as the rate of disclosure is small and does not offset the aging of the stock of undisclosed evidence. However, later in the purging phase, p falls sufficiently low that indifference can only be sustained with an increasing q; this requires κ to increase. Moreover, for q to approach its maximum value 1, it must be that the composition of the pool of undisclosed evidence shifts toward fresh evidence; this requires κ to tend to infinity as $t \to \overline{T}$. When the transparency phase begins, it is no longer possible to maintain indifference, and the sender discloses good evidence immediately as it arrives.

In Figure 3, it is noteworthy that the belief crosses $\underline{p}$ before the end of the purging phase. In other words, the sender continues to (stochastically) delay even though the receiver's belief p_t conditional on non-disclosure is sufficiently low that, had the game started at the same belief, there would be immediate disclosure at all times. The reason is that the incentive to delay depends not only on the current belief p_t but also the post-disclosure value q_t and its evolution; the presence of a stockpile of undisclosed evidence means that immediate disclosure of good evidence does not convince the receiver that the current state is 1, and delay can be weakly optimal. Thus, the variable q is a critical additional factor in the sender's best response problem.

4.1 Construction

Recall the definition of q_t as the receiver's belief immediately after disclosure of good evidence. Clearly, $q_0 = 1$. If $t \in (0, \underline{T})$, disclosure at time t is off-path, and passive belief updating means that the receiver's posterior beliefs over the timestamp is equal to the prior belief about the timestamp given that the evidence was good and there has been no disclosure before time t. Rather than maintain a belief over the whole interval of past times, it is convenient to introduce "stock" variables

$$G_{i,t} = \Pr(\tau \leq t \text{ and } \theta_\tau = 1 \text{ and } \theta_t = i | \text{no disclosure before } t) \qquad \text{for } i \in \{0, 1\},$$

representing the probability that the sender possesses good evidence *and* the state is currently i, conditional on no disclosure yet.

It follows that for $t \in (0, \underline{T})$,

$$q_t = \frac{G_{1,t}}{G_{1,t} + G_{0,t}}.$$

The same characterization also applies during the purging phase, since the disclosure rate of good evidence is independent of its timestamp.

With abuse of notation, let p denote the (deterministic) belief process conditional on no disclosure.

In either the stockpiling or purging phase, $(G_1, G_0, p^{\text{ND}})$ evolve according to

$$\begin{aligned}
\dot{G}_{1,t} &= -G_{1,t}\lambda_1 + (p_t - G_{1,t})\mu - G_{1,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \\
\dot{G}_{0,t} &= G_{1,t}\lambda_1 - G_{0,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \\
\dot{p^{\text{ND}}}_t &= -\lambda_1 p_t^{\text{ND}} - G_{1,t}\kappa_t + p_t^{\text{ND}}(G_{0,t} + G_{1,t})\kappa_t,
\end{aligned}$$

with initial values $(0, 0, p_0^{\text{ND}})$. In the G_1 and G_0 ODEs, the λ_1 terms reflect that Markov switching from state 1 to state 0 increases G_0 while decreasing G_1. The $(p_t - G_{1,t})\mu$ term represents the arrival of good evidence, which occurs at rate μ when the state is good and the sender does not already possess good evidence.[9] The $-\kappa_t G_{i,t}$ terms reflect outflows due to disclosure, and the adjustment factor $(1 - G_{1,t} - G_{0,t})$ accounts for the conditioning on no disclosure yet. The p-ODE terms have an analogous interpretation.[10]

In the stockpiling phase, $\kappa \equiv 0$, and the system has a closed form solution. The local incentive compatiblility condition for delay is

$$\dot{q}_t \geq r q_t - (\lambda_1 + r) p_t. \tag{2}$$

Intuitively, a marginal delay is beneficial when the current flow payoff p_t is large, the posterior after disclosing good evidence q_t is low, or when $\dot{q}_t$ is large (or not too negative). Provided $p_0 > \overline{p}$, the local delay IC condition holds until some time $\underline{T}$, when the sender becomes locally indifferent. This marks the end of the stockpiling phase.

Throughout the purging phase, the IC condition must hold with equality. The system in (G_1, G_0, p, q, κ) is a mixed differential-algebraic system rather than a standard initial value problem (IVP) since κ is not given exogenously, and instead we have an additional equation linking q to G_0 and G_1. To solve it, we eliminate G_0 and G_1 using $S := G_0 + G_1$ and $q = g_1/S$, and then we reduce the system to an IVP in two variables (p, q) by solving for S and κ as functions of (p, q). The dynamics of this system are illustrated in Figure 4. The lightly shaded, outlined region, R, is where S and κ are well defined and κ is nonnegative. In the proof, we show that (p, q) must exit this region at some finite time $\overline{T}$ — marking the end of the purging phase — and it does so through $q_{\overline{T}} = 1$. Hence, as $t \to \overline{T}$, the purging rate κ_t explodes and the stockpile empties, $S_t \to 0$.

After the purging phase, the stockpile is empty and the receiver conjectures immediate disclosure at all times. Hence, $q_t = 1$ for all $t \geq \overline{T}$. Since the receiver's belief p_t is below the threshold $\underline{p}$, immediate disclosure is optimal.

5 Discussion

5.1 Modeling Assumptions

Our model assumes the state evolves over time. Holding fixed the rest of the model, this assumption is in fact necessary for timestamps to play a nontrivial role; with a constant state,

[9]When we extend to the case $\lambda_0 > 0$, we must also consider the possibility that the sender already possesses bad evidence.

[10]When the bad state is not absorbing, we also must introduce analogous variables $B_{i,t}$ corresponding to bad evidence. The variable B_1 enters the law of motion for p when $\lambda_0 > 0$ since it is possible that the state is good but was bad in the past and the sender already obtained bad evidence, preventing the arrival of good evidence. When $\lambda_0 = 0$, B_1 is identically 0.

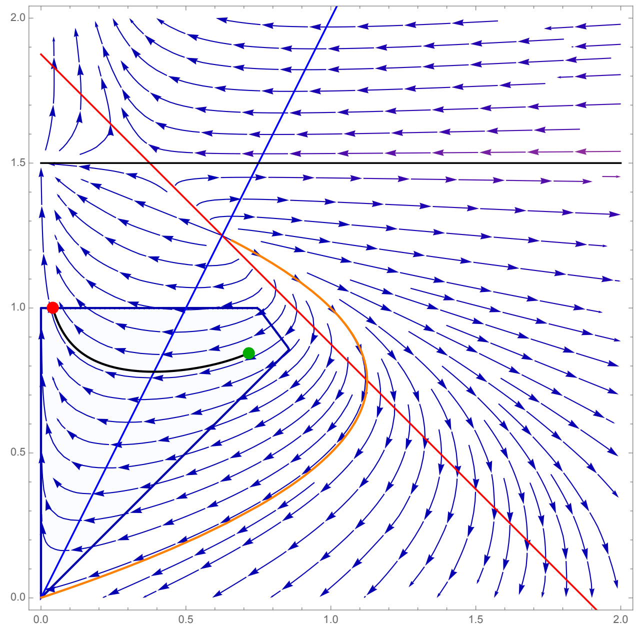

Figure 4: Phase diagram for reduced system in (p, q). The red and blue lines are isoclines for p and q, respectively, where their derivatives change sign. The ODEs are well defined below the black horizontal line. The orange curve is the stable manifold for the absorbing steady state $(0, 0)$. The solution first exits R through $q_{\overline{T}} = 1$.

the sender would immediately disclose good evidence and always disclose bad evidence, for all values of the other parameters.

We assume that evidence only arrives once. This is appropriate in settings where hard evidence is not continuously available, either because producing multiply pieces of evidence is infeasible or too costly. We conjecture that (i) delayed disclosure of good evidence and (ii) some disclosure of bad evidence would continue to be present in a richer model that allowed multiple arrivals of evidence. However, such a model would be more technically challenging as the receiver would have to maintain beliefs about a higher-dimensional evidence status of the sender.

We assume that the sender has no private information about the state other than the evidence she obtains. Since the sender's preferences are state-independent, equilibria in our model would also be equilibria in a model where the sender observes the state at all times. The latter model would allow for stronger off-path belief threats; for example, the receiver could assume that if good evidence is unexpectedly disclosed, the sender must know that the state is *currently* bad.

We assume that the only information available to the receiver (other than time itself and knowledge of the hidden Markov switching) is disclosure by the sender. If the model included exogenous public news, the evidence's timestamp would help the sender forecast continuation payoffs, changing the disclosure incentives.

5.2 Robustness and Other Equilibria

Proposition 3 uses passive belief updating off-path. With other off-path beliefs, a qualitatively similar result would obtain, but the delay region would have a different length; for example, if the receiver used a pessimistic off-path belief that good evidence was obtained at time 0, the stockpiling phase would be longer.

Although flexibility in off-path beliefs can produce multiplicity, the sender's expected payoff is the same in all equilibria. The following proposition establishes that as long as the receiver's conjecture is correct, the sender's expected payoff is the same, and thus the sender can also not benefit from commitment.[11]

Lemma 1. *Fix any $\lambda_1, \lambda_0, r, \mu > 0$. Fix any strategy of the sender, and let $(p_t)_{t\geq 0}$ denote the receiver's posterior belief process when that strategy is followed and the receiver's conjecture is correct. Then for all $t \geq 0$, $\mathbb{E}[p_t] = \phi_t = p^* + (p_0 - p^*)e^{-(\lambda_0+\lambda_1)t}$, which is independent of the sender's strategy.*

The proof is simple: by definition, p_t is the receiver's posterior expectation of the state θ_t, and by the law of iterated expectations, $\mathbb{E}[p_t] = \mathbb{E}[\theta_t] = \phi_t$, the receiver's belief about the state absent any information beyond the prior and the Markov switching dynamics.

Our restriction to $\lambda_0 = 0$ in the no-timestamps environment is useful as it isolates the analysis of good evidence disclosure by shutting down disclosure of bad evidence. That said, our results can be extended to $\lambda_0 > 0$. The following result gives a sufficient condition for there to continue to be an equilibrium with no disclosure of bad evidence.[12]

Proposition 4. *Assume $p_0 \in \left(\frac{\lambda_0}{\lambda_0+\lambda_1+r}, \frac{\lambda_0+r}{\lambda_0+\lambda_1+r}\right)$ and that $\mu < \left(\frac{\lambda_0+\lambda_1+r}{\lambda_1+r}\right)r$. Then there exists an equilibrium in which good evidence is disclosed immediately and bad evidence is never disclosed.*

The proposition gives a sufficient condition for no disclosure of bad evidence in the no-timestamps case. This gives a clean comparison between timestamps and no-timestamps. However, for very low prior beliefs, we conjecture that the receiver might disclose bad evidence immediately. This indicates that the role of timestamps on bad evidence is subtle: the threat that disclosure will be interpreted as bad could lead the sender to never disclose, or it could lead the sender to disclose sooner.

[11] This result holds for all $\lambda_1, \lambda_0 \geq 0$.

[12] Numerically, the equilibria in Proposition 3 are robust to small $\lambda_0 > 0$, and we conjecture that this can be proven by continuity with respect to λ_0 (in progress).

Apart from this result, within the no-timestamps environment, good and bad evidence have an interesting qualitative difference. With good evidence, if the receiver is conjecturing an atom of disclosure, the sender has an incentive to wait until just after the atom so that the evidence appears fresh. In this sense, the conjecture and the best reply are strategic substitutes. In contrast, with bad evidence, if the receiver conjectures an atom of disclosure, delay until just after the atom becomes *less* attractive, since fresh bad evidence is very harmful for beliefs. In this sense, a conjecture of fast disclosure of bad evidence can reinforce and incentive to disclose bad evidence early.

A Proofs

A.1 Proof of Proposition 1

The proof follows the following main steps. First, we show that in any equilibrium, good evidence is disclosed immediately (Lemma 2). Second, we show that bad evidence with any timestamp τ is disclosed after some (possibly infinite) time $D(\tau)$ increasing in τ (Corollary 1). Third, we characterize $D(\tau)$ uniquely.

Toward what follows, fix an arbitrary conjecture about the sender's strategy, and let p be the belief path conditional on no disclosure. Let $q_t^{G,\tau}$ and $q_t^{B,\tau}$ denote the belief after disclosure at time t of good and bad evidence, respectively, with timestamp τ. Note that $q_t^{G,\tau}$ is strictly increasing in τ, while $q_t^{B,\tau}$ is strictly decreasing in τ. Moreover, $q_t^{B,\tau}$ and $q_t^{G,\tau}$ are continuous in t and τ.

A.1.1 Immediate disclosure of good evidence

Lemma 2. *In any equilibrium, if the sender possesses good evidence at any time, she discloses it immediately.*

In the rest of this section, we prove Lemma 2.

Lemma 3. *If $p_t < q_t^{G,0}$ for all times, then immediate disclosure of good evidence is optimal regardless of its timestamp.*

Proof. Suppose the sender has good evidence with timestamp τ and hasn't disclosed it before time t. If the sender discloses at time $t' \geq t$, her flow payoff is p_s for $s \in [t, t')$ and $q_t^{G,\tau} \geq q_t^{G,0} > p_t$ for all $s \in [t', \infty)$. Immediate disclosure maximizes the flow payoff at all times. $\square$

Lemma 4. *If the sender possesses bad evidence at any time t and $p_t > q_t^{B,0}$, then there exists $\epsilon > 0$ such that the sender does not disclose it before time $t + \epsilon$.*

Proof. Suppose the sender has bad evidence with timestamp τ and hasn't disclosed it before time t. Since p and q^B are right-continuous, there exists $\epsilon > 0$ such that $p_s > q_s^{B,0} \geq q_s^{B,\tau}$ for all $s \in [t, t+\epsilon)$. Hence, by disclosing at time $t+\epsilon$ instead of t, the flow payoff is strictly higher on $[t, t+\epsilon)$ and is unchanged on $[t+\epsilon, \infty)$. □

Define $t^G = \inf\{t \geq 0 : p_t \geq q_t^{G,0}\}$ and $t^B = \inf\{t \geq 0 : p_t \leq q_t^{B,0}\}$. By right-continuity of p_t, $q_t^{G,0}$, and $q_t^{B,0}$ in t, if $p_0 \in (0,1)$, then $t^G, t^B > 0$. Right-continuity also implies $p_{t^G} \geq q_{t^G}^{G,0}$ if t^G is finite, and likewise $p_{t^B} \leq q_{t^B}^{B,0}$ if t^B is finite.

We now show that $t^G = \infty$. Clearly, since $q_t^{G,0} \neq q_t^{B,0}$ for all t, we cannot have $t^G = t^B < \infty$. Hence, if $t^G < \infty$, we have either $t^G < t^B$ or $t^B < t^G$.

The following lemma establishes that if t^G is finite, p must jump above $q^{G,0}$ from below $q^{B,0}$ at time t^G.

Lemma 5. *If $t^G < \infty$, then $t^B < t^G$ and* $\sup\{t < t^G : p_t \leq q_t^{B,0}\} = t^G$.

Proof. Toward a contradiction, suppose there exists $\epsilon > 0$ such that for all $t \in [t^G - \epsilon, t^G)$, $p_t > q_t^{G,0}$. By right-continuity, we can choose this ϵ such that $p_t > q_t^{G,0}$ for all $t \in [t^G - \epsilon, t^G + \epsilon)$. To simplify notation, set $t_0 := t^G - \epsilon$ and $t_1 := t^G + \epsilon$. Then by Lemma 4, there is no disclosure of bad evidence on $[t_0, t_1)$. Hence, on $[t_0, t_1)$, p is bounded above by the solution to $\dot{y}_t = -\lambda_1 y_t + \lambda_0(1 - y_t)$ with initial condition $y_{t_0} = p_{t_0}$. Here, y is the unconditional belief based on Markov switching, starting from p_{t_0}. To prove this, let $\mathbb{E}_t$ denote the receiver's expectation operator at time t, and $\mathbb{P}_t$ the probability measure at time t. By the law of iterated expectations, for all $t \in [t_0, t_1)$,

$$\mathbb{P}_{t_0}(\theta_t = 1) = \mathbb{P}_{t_0}(\text{disclosure by } t)\mathbb{P}_{t_0}(\theta_t = 1 | \text{disclosure by } t) + \mathbb{P}_{t_0}(\text{no disclosure by } t)p_t. \tag{3}$$

Since no bad evidence is disclosed in $[t_0, t_1)$, we have $\mathbb{P}_{t_0}(\theta_t = 1|\text{disclosure by } t) \geq q_t^{G,0}$. We also have

$$\mathbb{P}_{t_0}(\theta_t = 1) = y_t, \tag{4}$$

and

$$y_t < q_t^{G,0}.$$

The latter inequality holds because y and q^G satisfy the same linear ODE on $[t_0, t_1)$ and differ only in their values at the initial time; we have $q_{t_0}^G(0) > y_{t_0}$, so by the comparison theorem, $y_t < q_t^{G,0}$ for all $t \geq t_0$.

Putting these facts together, we have

$$\begin{aligned}
y_t &= \mathbb{P}_{t_0}(\text{disclosure by } t)\mathbb{P}_{t_0}(\theta_t = 1|\text{disclosure by } t) + \mathbb{P}_{t_0}(\text{no disclosure by } t)p_t \\
&\geq \mathbb{P}_{t_0}(\text{disclosure by } t)q_t^{G,0} + \mathbb{P}_{t_0}(\text{no disclosure by } t)p_t \\
&\geq \mathbb{P}_{t_0}(\text{disclosure by } t)y_t + \mathbb{P}_{t_0}(\text{no disclosure by } t)p_t.
\end{aligned}$$

Since $\mathbb{P}_{t_0}(\text{no disclosure by } t) > 0$, the last inequality implies that $p_t \leq y_t < q_t^{G,0}$ for all $t \in [t_0, t_1)$, contradicting the definition of t^G. □

Lemma 6. *We must have $t^G = \infty$.*

Proof. If $t^G < \infty$, then at time t, p jumps from weakly below $q^B(0)$ to weakly above $q^G(0)$. Such a jump requires that good or bad evidence is conjectured to be disclosed with an atom of probability at time t^G. Now if only an atom of good evidence disclosure is conjectured, p would exhibit a downward discontinuity, since any good evidence disclosure would raise the belief given that $p_{t^G-} \leq q^B_{t^G}(0) < q^G_{t^G}(0)$. Thus, at time t^G, the receiver must conjecture that bad evidence is disclosed with an atom of probability. But then disclosing bad evidence at time t^G would not be optimal, as there exists $\epsilon > 0$ such that $p_{t^G} > q^{B,0}_{t^G} \geq q^{B,\tau}_{t^G}$ for all $t \in [t^G, t^G + \epsilon)$ and all $\tau \in [0, t^G]$. In other words, the sender would strictly prefer to wait and disclose at time $t^G + \epsilon$. □

Since $t^G = \infty$, we have $p_t < q_t^{G,0}$ for all $t \geq 0$, and thus it is optimal to disclose good evidence immediately.

A.1.2 Delayed disclosure of bad evidence with timestamps

Lemma 7. *In equilibrium, p is continuous.*

Proof. If p has a discontinuity at some time t_0, then it must be that at time t_0, evidence is disclosed with an atom of probability. Since good evidence is disclosed immediately by Lemma 2 and arrives according to an atomless distribution, it must be that bad evidence is disclosed with an atom of probability at time t_0. Now for any type τ_0 with bad evidence that is willing to disclose at t_0, it must be that $q^{B,\tau_0}_{t_0} \geq p_{t_0}$, otherwise by right-continuity of p and continuity of q^{B,τ_0} we would have $p_t > q_t^{B,\tau_0}$ for all $t \in [t_0, t_0 + \epsilon)$ for some $\epsilon > 0$, and the sender would strictly prefer to disclose at $t_0 + \epsilon$.

We now argue that if there is a discontinuity at t_0, then $p_{t_0} < p_{t_0-}$, where the left limit exists because p is cadlag by Lemma **??**. For any $t < t_0$, and let N and G denote the events that there is no disclosure or there is good evidence disclosure in $[t, t_0]$ (suppressing dependence on t and t_0). Let B^{t_0} denote the event that there is bad evidence disclosure at time t_0, and $B^{[t,t_0)}$ the event that there is bad evidence disclosure in $[t, t_0)$. Let $\mathbb{E}_t$ and $\mathbb{P}_t$ denote expectations and probabilities conditional on no disclosure prior to t. By the law of iterated expectations,

$$\mathbb{E}_t[\theta_{t_0}] = \mathbb{P}_t(N)p_{t_0} + \mathbb{P}_t(G)\mathbb{E}_t[\theta_{t_0}|G] + \mathbb{P}_t(B^{[t,t_0)})\mathbb{E}_t[\theta_{t_0}|B^{[t,t_0)}] + \mathbb{P}_t(B^{t_0})\underbrace{\mathbb{E}_t[\theta_{t_0}|B^{t_0}]}_{\geq p_{t_0}},$$

where we can define $\mathbb{E}_t[\theta_{t_0}|B^{[t,t_0)}] = 0$ without loss in case $\mathbb{P}_t(B^{[t,t_0)}) = 0$. Now as $t \uparrow t_0$, $\mathbb{E}_t[\theta_{t_0}] \to p_{t_0-}$. Also, as $t \uparrow t_0$, we have $\mathbb{P}_t(G) \to 0$ and $\mathbb{P}_t(B^{[t,t_0)}) \to 0$, so $\mathbb{P}_t(N)+\mathbb{P}_t(B^{t_0}) \to 1$. Define $\alpha := \lim_{t\uparrow t_0} \mathbb{P}_t(N)$, $\beta := \lim_{t\uparrow t_0} \mathbb{P}_t(B^{t_0}) = 1 - \alpha$, and $\gamma := \lim_{t\uparrow t_0} \mathbb{E}_t[\theta_{t_0}|B^{t_0}] \geq p_{t_0}$. We have $\alpha > 0$ since the probability of no disclosure in $[t, t_0)$ is at least the probability that evidence arrives after time t_0, which is strictly positive, and we have $\beta > 0$ since there is an atom of disclosure at time t_0.

Thus, the limit of the right hand side (of our law of iterated expectations equation) as $t \uparrow t_0$ is $\alpha p_{t_0} + \beta\gamma \geq \alpha p_{t_0} + \beta p_{t_0} = p_{t_0}$. Since the left side has limit p_{t_0-}, it follows that $p_{t_0-} \geq p_{t_0}$. A discontinuity at t_0 must therefore imply that $p_{t_0-} > p_{t_0}$. This further implies $\gamma = \frac{p_{t_0-} - \alpha p_{t_0}}{\beta} > p_{t_0-}$.

Since $\lim_{t\uparrow t_0} \mathbb{E}_t[\theta_{t_0}|B^{t_0}] = \gamma > p_{t_0-}$, we have $\mathbb{E}_t[\theta_{t_0}|B^{t_0}] > p_{t_0-}$ for t sufficiently close to t_0. As $\mathbb{E}_t[\theta_{t_0}|B^{t_0}]$ is a convex combination over τ of $q_{t_0}^{B,\tau}$ values, there exists some timestamp $\tau < t_0$ for bad evidence for which the sender is willing to wait until t_0 to disclose and $q_{t_0}^{B,\tau} > p_{t_0-}$. By definition of p_{t_0-}, for any $\delta > 0$, there exists $\epsilon > 0$ such that if $t > t_0 - \epsilon$, $|p_t - p_{t_0-}| < \delta$. Choosing δ sufficiently small and ϵ accordingly, we have $p_t < q_t^{B,\tau}$ for all $t \in (t_0 - \epsilon, t_0)$. But then the sender would strictly prefer to disclose at any such t rather than wait until time t_0, contradicting optimality. $\square$

Lemma 8 (Single crossing)**.** *If $p_{t_0} \leq q_{t_0}^{B,\tau_0}$ for some $0 \leq \tau_0 \leq t_0$, then $p_t < q_t^{B,\tau_0}$ for all $t > t_0$.*

Proof. Assume $p_{t_0} \leq q_{t_0}^{B,\tau_0}$ for some $0 \leq \tau_0 \leq t_0$. Define $t^* = \inf\{t > t_0 : p_t \geq q_t^{B,\tau_0}\}$. We aim to show that $t^* = \infty$. First, we rule out $t^* \in (t_0, \infty)$, and then we rule out $t^* = t_0$. Toward a contradiction, suppose $t^* \in (t_0, \infty)$.

For each $\tau \leq t^*$, define $F(\tau) = \max_{t\in[\max\{t_0,\tau\},t^*]} \left(q_t^{B,\tau} - p_t\right)$. Since q^B is continuous in both arguments and p is continuous by Lemma 7, F is continuous by the maximum theorem. Moreover, F is strictly decreasing since q^B is strictly decreasing in τ and the set $[\max\{t_0, \tau\}, t^*]$ is weakly decreasing in τ in the set inclusion sense; it satisfies $F(\tau_0) > 0$ since $t^* > t_0$ by assumption, and by the definition of t^* we cannot have $p_t \geq q_t^{B,\tau_0}$ for any $t \in (t_0, t^*)$; and it satisfies $F(t^*) < 0$ since $q_{t^*}^B(t^*) = 0 < p_{t^*}$. Thus, there exists $\overline{\tau}$ such that $F(\tau) \geq 0$ if and only if $\tau \leq \overline{\tau}$. Since $F(\overline{\tau}) = 0$, there exists some $t_1 \in [\max\{t_0, \overline{\tau}\}, t^*)$ where

$q_{t_1}^{B,\overline{\tau}} = p_{t_1}$, with $q_t^{B,\overline{\tau}} \leq p_t$ for all $t \in [\max\{t_0, \overline{\tau}\}, t^*]$.

Define $\phi_t(p_{t_1}, t_1)$ as the solution to $\dot{y}_t = -\lambda_1 y_t + \lambda_0(1 - y_t)$ subject to the initial condition $y_{t_1} = p_{t_1}$. Note that q_t^{B,τ_0} solves the same ODE. Since $p_{t_1} < q_{t_1}^{B,\tau_0}$, it follows that $\phi_{t^*}(p_{t_1}, t_1) < q_{t^*}^{B,\tau_0} = p_{t^*}$. Now let N, G, and B denote the events that no evidence, good evidence, and bad evidence, respectively, is disclosed in $[t_1, t^*]$. By the law of iterated expectations,

$$\mathbb{E}_{t_1}[\theta_{t^*}] = \phi_{t^*}(p_{t_1}, t_1) = \mathbb{P}_{t_1}(N)p_{t^*} + \mathbb{P}_{t_1}(G)\mathbb{E}_{t_1}[\theta_{t^*}|G] + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B].$$

Now $\mathbb{E}_{t_1}[\theta_{t^*}|G] > q_{t^*}^{B,\tau_0} = p_{t^*}$. Hence,

$$\phi_{t^*}(p_{t_1}, t_1) \geq (1 - \mathbb{P}_{t_1}(B))p_{t^*} + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B].$$

Recall that $p_{t^*} > \phi_{t^*}(p_{t_1}, t_1)$, and $\mathbb{P}_{t_1}(B) < 1$, so we must have

$$\begin{aligned} &\phi_{t^*}(p_{t_1}, t_1) > (1 - \mathbb{P}_{t_1}(B))\phi_{t^*}(p_{t_1}, t_1) + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{t^*}|B] \\ \implies &\phi_{t^*}(p_{t_1}, t_1) > \mathbb{E}_{t_1}[\theta_{t^*}|B], \end{aligned}$$

where $\mathbb{P}_{t_1}(B) > 0$ as the first inequality cannot hold if $\mathbb{P}_{t_1}(B) = 0$. Since $\mathbb{E}_{t_1}[\theta_{t^*}|B] = \mathbb{E}_{t_1}[q_{t^*}^{B,\tau}|B]$, this means there must be some timestamp τ and some time $t \in (t_1, t^*)$ for which bad evidence is disclosed at time t and $q_{t^*}^{B,\tau} < \phi_{t^*}(p_{t_1}, t_1)$. Since $\phi_\cdot(p_{t_1}, t_1) = q_\cdot^{B,\overline{\tau}}$ on $[t_1, t^*]$ (as $q_\cdot^{B,\overline{\tau}}$ solves the same ODE with same initial value $q_{t_1}^{B,\overline{\tau}} = p_{t_1} = \phi_{t_1}(p_{t_1}, t_1)$), we have $q_{t^*}^{B,\tau} < q_{t^*}^{B,\overline{\tau}}$, and because $q_s^{B,\cdot}$ is strictly decreasing in the timestamp on $[0, s]$ for each fixed s, this implies $\tau > \overline{\tau}$. But then we have $F(\tau) < 0$, so $p_t > q_t^{B,\tau}$ for all $t \in (\max\{t_0, \tau\}, t^*)$, and this implies that the sender would strictly prefer to delay the disclosure until time t^*, a contradiction.

Next, we rule out the case $t^* = t_0$ by adapting arguments from the previous case. Toward a contradiction, suppose that $t^* = t_0$. Then for all $\epsilon > 0$ there exists $\hat{t} \in (t_0, t_0 + \epsilon)$ such that $p_{\hat{t}} \geq q_{\hat{t}}^{B,\tau_0}$. By arguments similar to those used in the previous case, there exists some timestamp $\overline{\tau} \geq \tau_0$ and time $t_1 \in [\max\{\overline{\tau}, t_0\}, \hat{t})$ such that $p_{t_1} = q_{t_1}^{B,\overline{\tau}}$ and $p_t \geq q_t^{B,\overline{\tau}}$ for all $t \in [\max\{\overline{\tau}, t_0\}, \hat{t}]$. Since $\phi_\cdot(p_{t_1}, t_1)$ and $q_\cdot^{B,\overline{\tau}}$ solve the same ODE and coincide at t_1, we have $\phi_\cdot(p_{t_1}, t_1) = q_\cdot^{B,\overline{\tau}}$ for all $s \geq t_1$, and in particular for $s = \hat{t}$.

Let N, G, and B be the events of no disclosure, good evidence disclosure, and bad evidence disclosure on $[t_1, \hat{t}]$. By similar logic to that used in the proof for the $t^* \in (t_0, \infty)$ case, we have $\phi_{\hat{t}}(p_{t_1}, t_1) = q_{\hat{t}}^{B,\overline{\tau}} \leq q_{\hat{t}}^{B,\tau_0} \leq p_{\hat{t}}$ and also $\mathbb{E}_{t_1}[\theta_{\hat{t}}|G] > q_{\hat{t}}^{B,\tau_0} \geq q_{\hat{t}}^{B,\overline{\tau}}$. Thus, by the law of iterated expectations,

$$\begin{aligned}\mathbb{E}_{t_1}[\theta_{\hat{t}}] = q_{\hat{t}}^{B,\overline{\tau}} &= \mathbb{P}_{t_1}(N)p_{\hat{t}} + \mathbb{P}_{t_1}(G)\mathbb{E}_{t_1}[\theta_{\hat{t}}|G] + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{\hat{t}}|B] \\ &> \mathbb{P}_{t_1}(N)q_{\hat{t}}^{B,\overline{\tau}} + \mathbb{P}_{t_1}(G)q_{\hat{t}}^{B,\overline{\tau}} + \mathbb{P}_{t_1}(B)\mathbb{E}_{t_1}[\theta_{\hat{t}}|B].\end{aligned}$$

As in the previous case, the strict inequality implies $\mathbb{P}_{t_1}(B) > 0$. Thus,

$$q_{\hat{t}}^{B,\overline{\tau}} > \mathbb{E}_{t_1}[\theta_{\hat{t}}|B].$$

Since the expectation is a convex combination of $q_{\hat{t}}^{B,\tau}$ values over $\tau \leq \hat{t}$, there must be disclosure at some time $t \in [t_1, \hat{t}]$ of bad evidence with some timestamp $\tau > \overline{\tau}$. But at t, we have $p_t \geq q_t^{B,\overline{\tau}} > q_t^{B,\tau}$. By continuity in t, this inequality holds for all times in an interval $[t, t+\epsilon)$ for some $\epsilon > 0$, and the sender would strictly prefer to delay disclosure until $t + \epsilon$, contradicting optimality. This rules out $t^* = t_0$. □

Corollary 1 (Deterministic delays)**.** *In equilibrium, bad evidence with timestamp τ is disclosed at some time $D(\tau) \in (\tau, +\infty]$. Moreover, there exists $\overline{\tau} \in (0, \infty]$ such that for $\tau \in [0, \overline{\tau})$, $D(\tau)$ is strictly increasing and finite, and for $\tau \geq \overline{\tau}$, $D(\tau) = +\infty$.*

Proof. Suppose that at some time t_0, the agent has bad evidence with timestamp $\tau_0 \leq t_0$. Let $D(\tau_0, t_0) := \inf\{t > t_0 : p_t \leq q_t^{B,\tau_0}\}$. (Note that $D(\tau_0, t_0) > \tau_0$, since $q_{\tau_0}^{B,\tau_0} = 0 < p_{\tau_0}$.) We claim that disclosing at time $D(\tau_0, t_0)$ is optimal. If $D(\tau_0, t_0) = +\infty$, then we have $p_t > q_t^{B,\tau_0}$ for all $t \geq t_0$, and clearly it is optimal for the agent to never disclose the evidence. If $D(\tau_0, t_0) < +\infty$, then we have (i) $p_t > q_t^{B,\tau_0}$ for all $t < D(\tau_0, t_0)$, which implies disclosing at time $D(\tau_0, t_0)$ yields a higher payoff than disclosing at any earlier time, and (ii) $p_t < q_t^{B,\tau_0}$ for all $t > D(\tau_0, t_0)$, which implies that disclosing at time $D(\tau_0, t_0)$ yields a higher payoff than disclosing at any later time. On path, it is therefore optimal to disclose at time $D(\tau) := D(\tau, \tau)$.

Define $\overline{\tau} := \inf\{\tau \geq 0 : D(\tau) = \infty\}$. Since $q_t^{B,\tau}$ is strictly increasing in t and strictly decreasing in τ, we have that $D(\tau)$ is increasing on $[0, \infty)$ and strictly increasing on $[0, \overline{\tau})$. □

We now turn to characterizing $\tau \mapsto D(\tau)$. Note that, in equilibrium, when $p_t = q_t^{B,s}$ for some s, p_t admits a direct characterization via Bayes' rule:

$$p_t = p_{t,s}^{\dagger} := \frac{e^{-\mu t}\phi_t + \int_s^t \mu e^{-\mu\tau}(1-\phi_\tau)q_t^{B,\tau}d\tau}{e^{-\mu t} + \int_s^t \mu e^{-\mu\tau}(1-\phi_\tau)d\tau}. \tag{5}$$

Here, for each s, $p_{t,s}^{\dagger}$ is an auxiliary belief process that specifies the probability that $\theta_t = 1$ conditional on no disclosure before time t, under a conjecture that (i) good evidence is always disclosed immediately and (ii) bad evidence with timestamp before s is disclosed before t, and no evidence with timestamp after s is disclosed before t. We will show that $p_{t^\dagger(s)} = p^\dagger_{t^\dagger(s),s}$ for all $s \geq 0$.[13]

Lemma 9. *If $\lambda_0, \lambda_1 > 0$, then for all $s \geq 0$, there exists $t^\dagger(s)$ that solves the following equation in t:*

$$p^\dagger_{t,s} = q_t^{B,s}. \tag{6}$$

Moreover, $t^\dagger(s)$ is unique and lies in $(s, +\infty)$.

Proof. Expanding (6) and multiplying through by the denominator yields

$$\begin{aligned} e^{-\mu t}\phi_t + \int_s^t \mu e^{-\mu\tau}(1-\phi_\tau) q_t^{B,\tau} d\tau &= q_t^{B,s}\left[e^{-\mu t} + \int_s^t \mu e^{-\mu\tau}(1-\phi_\tau)d\tau\right] \\ \iff \phi_t - q_t^{B,s} &= \int_s^t \mu e^{-\mu(\tau-t)}(1-\phi_\tau)(q_t^{B,s} - q_t^{B,\tau})d\tau. \end{aligned} \tag{7}$$

We establish existence by the intermediate value theorem. At $t = s$, the LHS is $\phi_s - 0 > 0$, while the RHS vanishes. As $t \to \infty$, the left hand side tends to $p^* - p^* = 0$. Meanwhile, for fixed $\epsilon > 0$, the right hand side is bounded below by

$$\int_{t-\epsilon}^t \mu e^{-\mu(\tau-t)}(1-\phi_\tau)(q_t^{B,s} - q_t^{B,\tau})d\tau \geq \int_{t-\epsilon}^t \mu e^{-\mu\epsilon}(1-\phi_\tau)(q_t^{B,s} - q_t^B(t-\epsilon))d\tau$$

As $t \to \infty$, we have $\phi_t \to p^* < 1$ under the assumption that $\lambda_1 > 0$, and we also have $q_t^{B,s} \to p^* > 0$ under the assumption that $\lambda_0 > 0$. In addition, $q_t^{B,t-\epsilon} < p^*(\lambda_0 + \lambda_1)\epsilon$. Hence, for small ϵ and large t the integrand is strictly positive, and we conclude that the RHS of (7) is positive for large t. Since the LHS and RHS of (7) are continuous in t, there exists a solution $t^\dagger(s) \in (s, +\infty)$ (and no solution at the ends of this interval).

Uniqueness follows from showing that $p^\dagger_{t,s}$ crosses $q_t^{B,s}$ from above at $t^\dagger(s)$. $\square$

A.2 Proof of Proposition 3

We first derive necessary conditions for optimality of the sender's strategy in a three-phase equilibrium and show that the prior belief determines the starting phase.

[13]For fixed s, $p^\dagger_{t,s}$ does not coincide at all times with the equilibrium belief process p_t, as these belief processes are based on different conjectures about the disclosure of bad evidence.

In the stockpiling phase, the laws of motion using $\lambda_0 = 0$ are

$$\begin{aligned}
\dot{G}_{1,t} &= -G_{1,t}\lambda_1 + (p_t - G_{1,t})\mu \\
\dot{G}_{0,t} &= G_{1,t}\lambda_1 \\
\dot{p}_t &= -\lambda_1 p_t \\
\dot{q}_t &= -\lambda_1 q_t + (1-q_t)\mu \frac{p_t - G_{1,t}}{S_t},
\end{aligned}$$

where $S_t := G_{1,t} + G_{0,t}$ is the total probability of having good evidence conditional on no disclosure.

This system admits a (unique) closed-form solution:

$$G_{1,t} = e^{-\lambda_1 t} p_0 (1 - e^{-\mu t}) \tag{8}$$

$$G_{0,t} = e^{-\lambda_1 t} p_0 \left(\frac{\lambda_1 e^{-t\mu} + \mu e^{\lambda_1 t}}{\mu + \lambda_1} - 1 \right) \tag{9}$$

$$p_t = e^{-\lambda_1 t} p_0 \tag{10}$$

$$q_t = \frac{G_{1,t}}{G_{0,t} + G_{1,t}} = \frac{(e^{\mu t} - 1)(\lambda_1 + \mu)}{(e^{(\lambda_1+\mu)t} - 1)\mu}. \tag{11}$$

A necessary and sufficient condition is that waiting until time $\underline{T}$ to disclose must be weakly preferred to disclosing at time t for a sender who obtains good evidence at time t. By disclosing immediately, the sender obtains a payoff of $\frac{q_t}{\lambda_1 + r}$. By disclosing at time $\underline{T}$, the sender obtains

$$\begin{aligned}
\int_t^{\underline{T}} e^{-r(s-t)} p_s \, ds &= \int_t^{\underline{T}} e^{-r(s-t)} p_t e^{-\lambda_1 (s-t)} \, ds + e^{-r(\underline{T}-t)} \frac{q_{\underline{T}}}{\lambda_1 + r} \\
&= \frac{p_t}{\lambda_1 + r}(1 - e^{-(\lambda_1+r)(\underline{T}-t)}) + e^{-r(\underline{T}-t)} \frac{q_{\underline{T}}}{\lambda_1 + r},
\end{aligned}$$

where we have used that p_s is calculated under the conjecture that no evidence is disclosed before time $\underline{T}$.

Hence, by comparing these payoffs, a necessary and sufficient condition is

$$q_t \leq p_t(1 - e^{-(\lambda_1+r)(\underline{T}-t)}) + e^{-r(\underline{T}-t)} q_{\underline{T}} \qquad \text{for all } t \leq \underline{T}.$$

Since $p_t e^{-\lambda_1(\underline{T}-t)} = p_{\underline{T}}$, this rearranges to

$$e^{-rt}(q_t - p_t) \leq e^{-r\underline{T}}(q_{\underline{T}} - p_{\underline{T}}) \qquad \text{for all } t \leq \underline{T}.$$

A necessary condition is then

$$0 \leq \frac{d}{dt}\left[e^{-rt}(q_t - p_t)\right]\Big|_{t=\underline{T}} = e^{-rt}[-r(q_t - p_t) + \dot{q}_t - \dot{p}_t)\Big|_{t=\underline{T}}$$
$$= e^{-rt}[\dot{q}_t - rq_t + (\lambda_1 + r)p_t\Big|_{t=\underline{T}}$$

for some $\underline{T} > 0$.

This condition implies a further necessary condition that, for some $t > 0$,

$$p_0 \geq b(t) := \frac{e^{\lambda_1 t}(rq_t - \dot{q}_t)}{\lambda_1 + r}.$$

We claim that $b'(t) > 0$ for all $t > 0$. Direct calculation shows that

$$b'(t) = \frac{e^{\lambda_1 t}(\lambda_1 + \mu)}{(e^{(\lambda_1+\mu)t} - 1)^3(\lambda_1 + r)\mu} P(t), \qquad \text{where}$$
$$P(t) := \mu(\lambda_1 + \mu + r)\, e^{t(2\lambda_1 + 2\mu)} - (\lambda_1 + \mu)(2\lambda_1 + \mu + r)\, e^{t(\lambda_1 + 2\mu)}$$
$$+ (2\lambda_1^2 + 3\lambda_1\mu + \lambda_1 r + \mu^2 - \mu r)\, e^{t(\lambda_1+\mu)} - (\lambda_1 + \mu)(\mu - r)\, e^{t\mu} - \lambda_1 r].$$

Note that for both $\mu \geq r$ and $\mu < r$, $P(t)$ has three sign changes and thus at most three real roots. Moreover, it is easy to check that there is a triple root at $t = 0$. Hence, there are no other real roots. Also, since the leading term is positive, $P(t)$ is positive for sufficiently large t. We conclude that for all $t > 0$, we have $P(t) > 0$ and therefore $b'(t) > 0$. It follows that $p_0 \geq b(t)$ for some $t > 0$ only if

$$p_0 > b(0) = \frac{r \cdot q_0 - \dot{q}_0}{\lambda_1 + r} = \frac{r \cdot 1 + \frac{\lambda_1}{2}}{\lambda_1 + r}.$$

For later use, let us define $c(t) = \dot{q}_t - rq_t + (\lambda_1 + r)p_t$. Assume $p_0 > \overline{p}$. We show that c crosses 0 from above at the unique point t where $p_0 = b(t)$. We have

$$c(t) = (\lambda_1 + r)e^{-\lambda_1 t}(p_0 - b(t))$$
$$\implies c'(t) = -\lambda_1 c(t) - b'(t)(\lambda_1 + r)e^{-\lambda_1 t}.$$

When $c(t) = 0$, the right hand side has the same sign as $-b'(t) < 0$, as desired.

Next, we derive a necessary condition for immediate disclosure to be optimal over a positive-measure interval of time starting at the beginning of the game; this gives a necessary condition for a three-phase equilibrium to only feature the transparency phase. Suppose the receiver conjectures immediate disclosure over an interval $[0, T)$ for $T > 0$. Then trivially, we have $q_t = 1$ for all $t \in [0, T)$. Now suppose the sender obtains good evidence at time 0 and discloses at time $t \in [0, T)$. Then for $s \geq t$, we have $p_s = e^{-\lambda_1(s-t)}$. The sender's payoff is

$$F(t) := \int_0^t e^{-rs} p_s \, ds + \int_t^\infty e^{-rs} q_t e^{-\lambda_1(s-t)} \, ds.$$

A necessary condition is that $F(t)$ has a local maximum at $t = 0$:

$$0 \geq F'(0) = p_0 - q_0 + \int_t^\infty (\lambda_1 q_t + \dot{q}_t) e^{-rs} e^{-\lambda_1(s-t)} \, ds \Big|_{t=0}. \tag{12}$$

Since $q_t = 1$ for $t \in [0, T)$, the right hand side evaluates to

$$p_0 - 1 + \lambda_1 \frac{1}{\lambda_1 + r}.$$

If this is nonnegative, then it must be that

$$p_0 \leq \frac{r}{\lambda_1 + r}.$$

Finally, to derive a necessary condition for the purging phase to begin at time 0, note that the sender must be indifferent between disclosing good evidence at time 0 and disclosing at time t for all sufficiently small $t > 0$. Hence, for all sufficiently small $t > 0$,

$$\begin{aligned} & 0 = F'(t) \\ \iff & 0 = p_t - q_t + (\lambda_1 q_t + \dot{q}_t) \frac{1}{\lambda_1 + r} \\ \iff & \dot{q}_t = r q_t - (\lambda_1 + r) p_t. \end{aligned} \tag{13}$$

At $t = 0$, we have $q_0 = 1$ and $\dot{q}_0 \leq 0$ as q is bounded above by 1. This implies $p_0 \geq \frac{r}{\lambda_1 + r}$. Moreover, we can rule out $p_0 = \frac{r}{\lambda_1 + r}$ as follows. Note that this would require $\dot{q}_0 = 0$. Since q is bounded above by 1, we have $\ddot{q}_0 \leq 0$. If indifference holds for all $t \in [0, \epsilon)$, then $\ddot{q}_0 = r\dot{q}_0 - (\lambda_1 + r)\dot{p}_0 = 0 - (\lambda_1 + r)\dot{p}_0 \geq (\lambda_1 + r)\lambda_1 p_0 > 0$, where the bound $\dot{p}_0 \leq -\lambda_1 p_0$ follows from $p_t \leq \phi_t = \phi_0 e^{-\lambda_1 t}$. This contradicts $\ddot{q}_0 \leq 0$, so we conclude that the game can begin with a purging phase only if $p_0 > \frac{r}{\lambda_1 + r}$.

We argue that $\dot{q}_0 \geq -\frac{\lambda_1}{2}$. Let $X_t = t - \tau$ be the age of evidence that is disclosed at time t. Then

$$\dot{q}_0 = -\lambda_1 \lim_{t \to 0} \frac{\mathbb{E}[X_t]}{t}.$$

Expanding and writing $\gamma_s := \phi_s \mu e^{-\mu s}$ yields

$$\lim_{t\to 0}\frac{\mathbb{E}[X_t]}{t} = \lim_{t\to 0}\frac{\int_0^t (t-s)\gamma_s e^{-\int_s^t \kappa_u\,du}\,ds}{t\int_0^t \gamma_s e^{-\int_s^t \kappa_u\,du}\,ds} = \lim_{t\to 0}\frac{\int_0^t (t-s)e^{-\int_s^t \kappa_u\,du}\,ds}{t\int_0^t e^{-\int_s^t \kappa_u\,du}\,ds},$$

where we have used that γ is continuous. Since the density $\frac{e^{-\int_s^t \kappa_u\,du}}{\int_0^t e^{-\int_x^t \kappa_u\,du}\,dx}$ is nondecreasing in s, while the function $t-s$ is decreasing, the weighted average is bounded above by replacing the density with a uniform density. Thus,

$$\lim_{t\to 0}\frac{\mathbb{E}[X_t]}{t} \le \lim_{t\to 0}\frac{\int_0^t (t-s)\,ds}{t\int_0^t \,ds} = \lim_{t\to 0}\frac{\frac{t^2}{2}}{t^2} = \frac{1}{2}.$$

We conclude that $\dot{q}_0 \ge -\frac{\lambda_1}{2}$, and thus

$$p_0 = \frac{r-\dot{q}_0}{\lambda_1 + r} \le \frac{r+\frac{\lambda_1}{2}}{\lambda_1 + r}.$$

We now establish existence and uniqueness within the class of three-phase equilibria for p_0 in each of the three intervals.

A.2.1 Low p_0.

Suppose $p_0 \le \frac{r}{\lambda_1 + r}$ and that the receiver conjectures that bad evidence is never disclosed, while good evidence is disclosed immediately. Then $q_t \equiv 1$ and p evolves according to

$$\dot{p}_t = -\lambda_1 p_t - \mu p_t(1-p_t).$$

Note that $\dot{p}$ is strictly decreasing. Suppose the sender has good evidence at time t_0; without loss, set $t_0 = 0$. By waiting until time s, the sender obtains

$$F(t) = \int_0^t e^{-rs} p_s\,ds + \int_t^\infty e^{-rs} 1 e^{-\lambda_1(s-t)}\,ds,$$

which has derivative

$$\begin{aligned}
F'(t) &= p_t - 1 + \int_t^{\infty} \lambda_1 e^{-rs} e^{-\lambda_1 (s-t)}\, ds \\
&= p_t - 1 + e^{-rt} \frac{\lambda_1}{\lambda_1 + r} \\
&< p_t - \frac{r}{\lambda_1 + r} \\
&< 0,
\end{aligned}$$

where we have used that p is strictly decreasing and $p_0 < \frac{r}{\lambda_1 + r}$.

Hence, immediate disclosure of evidence is optimal at all histories, including after off-path delay of disclosure.

A.2.2 High p_0.

Suppose that $p_0 > \frac{r+\lambda_1/2}{r+\lambda_1}$. Then any three-phase equilibrium must start in the stockpiling phase, with $0 < \underline{T} < \overline{T} < \infty$. Suppose the receiver conjectures that the sender follows a strategy as in the proposition, characterized by some $\underline{T}_0$, $\overline{T}_0$, and $(\kappa_t)_{t \in [\underline{T}_0, \overline{T}_0)}$. We derive necessary conditions that uniquely determine these objects and then verify that the sender's strategy is optimal.

For $t \in [0, \underline{T}_0)$, it must be that the sender prefers to disclose at $\underline{T}_0$ rather than at t. From the earlier step in the proof where we derived the lower bound $\overline{p}$ for there to be a nonempty delay region, we must have

$$\dot{q}_{\underline{T}_0} \geq r q_{\underline{T}_0} - (\lambda_1 + r) p_{\underline{T}_0} \tag{14}$$

There is a unique value, which we denote $\underline{T}$, where this condition holds with equality, and it holds with strict inequality at each $t \in [0, \underline{T})$. From the earlier analysis, equality holds if and only if $p_0 = b(t)$, where we showed that $b'(t) > 0$ for all $t > 0$, and by definition, $p > b(0)$. Moreover, it is easy to show that $\lim_{t \to 0} b(t) > 1 > p_0$. Hence, there is a unique crossing point, $\underline{T}$.

We claim that $\underline{T}$ is the only candidate for the end of the stockpiling phase. Clearly, we cannot have an equilibrium with $\underline{T}_0 > \underline{T}$, since the incentive condition for delay would fail at $\underline{T}$. We will rule out $\underline{T}_0 < \underline{T}$ as part of our analysis of the purging phase.

In the purging phase, $[\underline{T}_0, \overline{T}_0)$, the local indifference condition (13) must hold. Moreover,

(p, G_1, G_0) have the following laws of motion:

$$\dot{G}_{1,t} = -G_{1,t}\lambda_1 + (p_t - G_{1,t})\mu - G_{1,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \tag{15}$$

$$\dot{G}_{0,t} = G_{1,t}\lambda_1 - G_{0,t}\kappa_t(1 - G_{1,t} - G_{0,t}) \tag{16}$$

$$\dot{p}_t = -\lambda_1 p_t - G_{1,t}\kappa_t + p_t(G_{0,t} + G_{1,t})\kappa_t, \tag{17}$$

with initial values $(p_{\underline{T}_0}, G_{1,\underline{T}_0}, G_{0,\underline{T}_0})$ determined by continuity at the end of the delay region. Note that q must also satisfy, by definition, $q_t = G_{1,t}/(G_{1,t} + G_{0,t})$ at each t in $[\underline{T}_0, \overline{T}_0)$.[14]

To solve the system, first, use the change of variables $S = G_0 + G_1$ and $q = G_1/S$ and differentiate to obtain

$$\dot{S}_t = (p_t - S_t q_t)\mu - S_t\kappa_t(1 - S_t) \tag{18}$$

$$\dot{q}_t = -\lambda_1 q_t + (1 - q_t)\mu\frac{p_t - S_t q_t}{S_t} \tag{19}$$

Using the indifference equation (13), we can eliminate $\dot{q}_t$ and solve for S_t:

$$\dot{q}_t = -\lambda_1 q_t + (1 - q_t)\mu\frac{p_t - S_t q_t}{S_t} = rq_t - (\lambda_1 + r)p_t \tag{20}$$

$$\implies S_t = S(p_t, q_t) := \frac{\mu p_t(1 - q_t)}{q_t[\lambda_1 + r + \mu(1 - q_t)] - (\lambda_1 + r)p_t}. \tag{21}$$

A sufficient condition for S_t to be well defined is that $p_t < q_t \leq 1$, which we will verify.

To solve for κ as a function of other variables, differentiate (21), substitute in the expressions for $\dot{p}$ and $\dot{q}$ from (17) and (13), and equate the resulting expression to the right hand side of (18). Solving for κ_t yields

$$\kappa_t = \kappa(p_t, q_t) := \frac{A_t(q_t[\lambda_1 + r + \mu(1 - q_t)] - (\lambda_1 + r)p_t)}{(q_t - p_t)(1 - q_t)[\lambda_1 + r + \mu(1 - q_t)]}, \qquad \text{where} \tag{22}$$

$$A_t := A(p_t, q_t) := r + \lambda_1 + \mu + (r - \mu)q_t - 2p_t(\lambda_1 + r). \tag{23}$$

As long as $0 < p_t < q_t < 1$, which we will verify in our solution, denominator will be positive, so κ_t will be well defined. We will also need to verify that $\kappa_t \geq 0$ at all times for it to be feasible as an arrival rate. Since $0 < p_t < q_t < 1$ implies the second factor in the numerator is positive, it will be sufficient to show that $A_t \geq 0$ for all $t \in [\underline{T}_0, \overline{T}_0)$.

After eliminating S_t and κ_t in this way, we obtain the reduced IVP consisting of the pair

[14] Since $\underline{T}_0 > 0$, $G_{1,\underline{T}_0}/(G_{1,\underline{T}_0} + G_{0,\underline{T}_0})$ is well defined.

of ODEs

$$\dot{p}_t = p_t \frac{2(\lambda_1 + r)\mu p_t + \mu q_t(\lambda_1 + \mu - r) - (\lambda_1 + \mu)(\lambda_1 + r + \mu)}{\lambda_1 + r + \mu(1 - q_t)} \tag{24}$$

$$\dot{q}_t = r q_t - (\lambda_1 + r) p_t, \tag{25}$$

with initial conditions $(p_{\underline{T}_0}, q_{\underline{T}_0})$ from the end of the stockpiling phase.

Note that the right hand sides are time-invariant, C^1 functions of (p_t, q_t) on the domain $U := \{(p, q) \in \mathbb{R}^2 : q < 1 + \frac{\lambda + r}{\mu}\}$. In particular, they are locally Lipschitz continuous, uniformly in time. By the Picard-Lindelöf theorem, there exists a unique solution to the IVP with initial conditions on some interval $[\underline{T}_0, \underline{T}_0 + \epsilon]$ with $\epsilon > 0$. Let $[\underline{T}, T^{\max})$ denote the maximal interval of existence (and uniqueness), where $T^{\max}$ is possibly $+\infty$.

Let $R \subset \mathbb{R}^2$ denote the region for which (i) κ_t is well defined via (22) and strictly positive and (ii) S_t is well defined via (21):

$$R := \{(p, q) : 0 < p < q < 1 \text{ and } A(p, q) > 0\} \subset U.$$

Define the (possibly infinite) time $T := \inf\{t \in [\underline{T}_0, T^{\max}) : (p_t, q_t) \notin R\}$.

Lemma 10. *We have that $(p_{\underline{T}_0}, q_{\underline{T}_0})$ lies in the interior of R. Hence, $T > \underline{T}_0$.*

Proof. In the stockpiling phase, we have

$$\dot{p}_t = -\lambda_1 p_t \tag{26}$$

$$\dot{q}_t = -\lambda_1 q_t + (1 - q_t)\mu \frac{p_t - S_t q_t}{S_t} \tag{27}$$

$$\dot{S}_t = (p_t - S_t q_t)\mu, \tag{28}$$

with initial conditions $(p_0, q_0, S_0) = (p_0, 1, 0)$. That p_t is strictly positive for all $t \in [0, \underline{T}_0]$ is standard; this follows from the comparison theorem and is also evident from the closed form solution.

It is useful to show that $S_t > 0$ until $\underline{T}_0$. We have $\dot{S}_0 > 0$, so $S_t > 0$ for all sufficiently small t. Now write (28) as $\dot{S}_t = f(t, S_t)$, and define $y_t \equiv 0$. Then $S_t > y_t$ and $\dot{S}_t - f(t, S_t) = 0 > -p_t\mu = y_t - f(t, y_t)$. By the comparison theorem implies $S_t > y_t = 0$ for all $t > 0$.

The comparison theorem also implies $q_t > 0$ for all $t \in [0, \underline{T}]$; moreover, $q_t < 1$ for all $t \in (0, \underline{T}]$, since $\dot{q}_0 = -\lambda_1 < 0$ and the RHS of the q-ODE is negative whenever $q_t = 1$.

Also note that $N_t := p_t - S_t q_t = p_t - G_{1,t}$, which has ODE $\dot{N}_t = -(\lambda_1 + \mu)N_t$. Since $N_0 = p_0 - G_{1,0} = p_0 \in (0, 1)$, we have $N_t = p_t - G_{1,t} > 0$ for all $t \in [0, \underline{T})$.

Define $\delta_t := q_t - p_t$, where $\delta_0 = 1 - p_0 > 0$. We have

$$\dot{\delta}_t = -\lambda_1 \delta_t + (1 - q_t)\mu \frac{N_t}{S_t}. \tag{29}$$

The inequalities just established imply the second term is positive for all $t \in (0, \underline{T}]$. Hence, by the comparison theorem, we have $\delta_t > 0$ for all $t \in [0, \underline{T}]$.

To prove that $A_{\underline{T}_0} > 0$, recall from earlier in the proof that $c'(\underline{T}) < 0$. Hence, $c(\underline{T})$ crosses 0 from above at $\underline{T}$. Now $c(\underline{T}) = 0$ and (27) imply

$$0 = \dot{q}_t - r q_t + (\lambda_1 + r) p_t = \dot{q}_t + \lambda_1 q_t - (1 - q_t)\mu \frac{p_t - S_t q_t}{S_t} \tag{30}$$

$$\implies S_t = S(p_t, q_t) \tag{31}$$

Now when $c(t) = 0$, some algebra shows that $c'(t) = -\frac{(\lambda_1 + r)(q_t - p_t)}{1 - q_t} A_t$. Since $c(\underline{T}) = 0$ by construction and $c'(\underline{T}) < 0$, it follows that $A_{\underline{T}} > 0$. □

Lemma 11. *If $T < T^{\max}$, then $q_T > p_T > 0$ and $A_T > 0$. Hence, $q_T = 1$.*

Proof. Throughout the proof, suppose $T < T^{\max}$. Since the right hand side of (24) contains a factor of p_t, the comparison theorem implies $p_t > 0$ for all $t \in [\underline{T}, T^{\max})$, and in particular, $p_T > 0$.

Note that for all $t \in [\underline{T}, T)$, we have $(p_t, q_t) \in R$ and therefore κ_t is well-defined and strictly positive. Moreover, S_t is well-defined and strictly positive. Hence, by construction, the ODEs for (G_1, G_0, p) are all satisfied.

The original ODE for p can be written

$$\dot{p}_t = -\lambda_1 p_t - S_t \kappa_t (q_t - p_t). \tag{32}$$

Defining $\delta_t = q_t - p_t$ and subtracting this ODE for p from the indifference condition for q, we have

$$\dot{\delta}_t = \delta_t (r + S_t \kappa_t). \tag{33}$$

It follows that δ is strictly increasing on $[\underline{T}, T)$, so $\delta_T > 0$; in other words, $q_T > p_T$.

We now establish that $A_T > 0$. Since $A_t > 0$ for all $t < T$, we must have $A_T \geq 0$. Toward a contradiction, suppose $A_T = 0$. We claim that $\dot{A}_T > 0$. A_t is differentiated as

$$\begin{aligned}\dot{A}_t &= (r-\mu)\dot{q}_t - 2p_t(\lambda_1 + r)\\ &= (r-\mu)[rq_t - (\lambda_1+r)p_t]\\ &\quad - 2(\lambda_1+r)\left[p_t \frac{2(\lambda_1+r)\mu p_t + \mu q_t(\lambda_1+\mu - r(\lambda_1+\mu)(\lambda_1+r+\mu)}{\lambda_1+r+\mu(1-q_t)}\right].\end{aligned}$$

Now if $A_t = 0$, then $p_t = \frac{r+\lambda_1+\mu+(r-\mu)q_t}{2(\lambda_1+r)}$. Substituting this into the expression for $\dot{A}_t$ and simplifying yields

$$\begin{aligned}\dot{A}_t|_{A_t=0} &= C_0 + C_1 q_t, \qquad \text{where}\\ C_0 &:= \frac{1}{2}(2\lambda_1+\mu-r)(\lambda_1+r+\mu)\\ C_1 &:= \frac{1}{2}(r-\mu)(2\lambda_1+\mu+r).\end{aligned}$$

Since this is linear in q_t, it suffices to sign at extreme values. Using $q_t = 1$ yields $C_0 + C_1 = \frac{1}{2}\lambda_1(3r + 2\lambda_1 + \mu) > 0$. To evaluate at a lower bound on q_t, recall that $A_T = 0$ implies

$$p_T = \frac{r + \lambda_+\mu + (r-\mu)q_T}{2(\lambda_1+r)} < q_T,$$

as we have already shown $p_T < q_T$. This in turn yields $q_T > \underline{q} := \frac{\lambda_1+\mu+r}{2\lambda_1+\mu+r}$. Hence,

$$\begin{aligned}C_0 + C_1\underline{q} &= \frac{1}{2}(2\lambda_1+\mu-r)(\lambda_1+r+\mu) + \frac{1}{2}(r-\mu)(\lambda_1+\mu+r)\\ &= \lambda_1(\lambda_1+r+\mu) > 0.\end{aligned}$$

As $q_t \in (\underline{q}, 1)$ it follows that $C_0 + C_1 q_t$ lies between two positive values obtained at the extreme values $\underline{q}$ and 1, and is therefore positive.

We conclude that $\dot{A}_T|_{A_T=0} > 0$, so A_t cannot intersect 0 from above at $t = T$. Thus, $A_T > 0$.

Now the 0 function solves the p-ODE, so by the comparison theorem, $p_t > 0$ for all $t \in [\underline{T}, T)$.

Having established that $q_T > p_T > 0$ and $A_T > 0$, the only remaining possibility is that $q_T = 1$. □

We now show that indeed $T < T^{\max}$.

There are two alternative possibilities: either $T = T^{\max} < \infty$ or $T = T^{\max} = \infty$. The following lemma rules out the first possibility.

Lemma 12. *If $T = T^{\max}$, then both are infinite.*

Proof. If $T = T^{\max}$ and T is finite, then we have $q_T = 1$, so (p_T, q_T) is in the interior of U by its definition. This implies that the solution extends uniquely beyond T, so $T^{\max} > T$, a contradiction. □

Lemma 13. *It does not hold that $T = T^{\max} = \infty$.*

Proof. Suppose $T = T^{\max} = \infty$. Then a unique solution exists for all time and remains in R. But the ODE for $\delta = p - q$ has solution

$$\delta_t = \delta_{\underline{T}} e^{\int_{\underline{T}}^{t} (r + S_s \kappa_s)\, ds} \geq \delta_{\underline{T}} e^{r(t - \underline{T})}$$

where we have used that $S_t > 0$ and $\kappa_t > 0$ for all t, since $(p_t, q_t) \in R$. In particular, δ reaches 1 in finite time. This contradicts the fact that in R, $0 < p_t < q_t < 1$. □

To summarize, $T < T^{\max}$, and the unique maximal solution exits R at time T with $q_T = 1$. It follows that as $t \to T$, $\kappa_t \to \infty$ and $S_t \to 1$. Standard verification shows that the sender's strategy is sequentially rational.

A.2.3 Moderate p_0 (in progress)

Suppose $p_0 \in (\underline{p}, \overline{p}]$. In this case, the equilibrium must begin in the purging phase. The system is the same as in the purging phase for the high-p_0 case, except for minor technical modifications due to the initial conditions $(G_0, G_1, S_0, q_0) = (0, 0, 0, 1)$. This means that κ_t given by (22) is not finite at time 0, and the right side of (19) is not well defined at time 0. Nonetheless, there is a unique solution of the reduced system in (p, q), S is well defined by (21), and κ is well defined by (22) except at time 0. There is a unique pair (G_1, G_0) that is continuous at time 0, satisfies the initial conditions, and satisfies (15)-(16) at all $t > 0$. Moreover, (17), (18), and (19) are satisfied at all $t > 0$ by construction.

A.3 Proof of Proposition 4

We verify optimality of the sender's strategy assuming the receiver correctly conjectures immediate disclosure of good evidence and no disclosure of bad evidence, and that following off-path disclosure of bad evidence at time t, the receiver assumes the evidence arrived at time t.

Let $q_{B,t}$ (identically 0 under our off-path belief assumption) denote the receiver's posterior immediately after disclosure of bad evidence.

Some algebra shows that delaying bad evidence disclosure is locally incentive compatible if and only if

$$\dot{q}_{B,t} \geq (\lambda_0 + \lambda_1 + r)(q_{B,t} - p_t) + (\lambda_0 + \lambda_1)(p^* - q_{B,t}).$$

With $q_B \equiv 0$, this reduces to

$$p_t \geq \frac{\lambda_0}{\lambda_0 + \lambda_1 + r}, \tag{34}$$

and hence to establish that it is optimal to never disclose bad evidence, it suffices to show that the receiver's belief conditional always satisfies this inequality. Note that by assumption, the inequality is satisfied strictly at time 0.

Let $q_{G,t}$ (identically 1 under the conjectured strategy) denote the receiver's belief immediately after disclosure of good evidence. Delaying disclosure of good evidence is not locally incentive compatible if

$$\begin{aligned}\dot{q}_{G,t} &< (\lambda_0 + \lambda_1 + r)(q_{G,t} - p_t) + (\lambda_0 + \lambda_1)(p^{**} - q_{G,t}) \\ &= r q_{G,t} - (\lambda_0 + \lambda_1 + r)p_t + \lambda_0.\end{aligned}$$

With $q_G \equiv 1$, this reduces to

$$p_t < \frac{\lambda_0 + r}{\lambda_0 + \lambda_1 + r}. \tag{35}$$

Hence, whenever this inequality is satisfied, the sender prefers to disclose good evidence immediately rather than at any future time. Recall that this inequality is satisfied at time 0 by assumption.

We now show that under the condition on μ in the proposition, $p_t \in \left(\frac{\lambda_0}{\lambda_0+\lambda_1+r}, \frac{\lambda_0+r}{\lambda_0+\lambda_1+r}\right)$ for all $t \geq 0$ conditional on no disclosure.

Let $B_{x,t}$ denote the probability that the sender has bad evidence and $\theta_t = x \in \{0, 1\}$, conditional on no disclosure yet. The laws of motion for (B_1, B_0, p) under the receiver's conjecture are

$$\dot{B}_{1,t} = B_{0,t}\lambda_0 - B_{1,t}\lambda_1 + B_{1,t}(p_t - B_{1,t})\mu. \tag{36}$$

$$\dot{B}_{0,t} = -B_{0,t}\lambda_0 + B_{1,t}\lambda_1 + (1 - p_t - B_{0,t})\mu + B_{0,t}(p_t - B_{1,t})\mu \tag{37}$$

$$\dot{p}_t = \lambda_0(1 - p_t) - \lambda_1 p_t - \mu(p_t - B_{1,t})(1 - p_t). \tag{38}$$

with $B_{0,0} = B_{1,0} = 0$.

Standard arguments using the comparison theorem show that there is a unique solution (B_1, B_0, p) to this system for all time, all variables lie in $(0,1)$, and $p_t > B_{1,t}$. By inspection, the right hand side of (38) is negative when p_t exceeds $p^* = \frac{\lambda_0}{\lambda_0+\lambda_1}$, which lies in $\left(\frac{\lambda_0}{\lambda_0+\lambda_1+r}, \frac{\lambda_0+r}{\lambda_0+\lambda_1+r}\right)$. Hence, p_t cannot hit the upper end of this interval in finite time.

Next, note that the right hand side of (38) is bounded below by

$$f(p_t) := \lambda_0(1-p_t) - \lambda_1 p_t - \mu(p_t - 0)(1-p_t).$$

Let $\tilde{p}$ denote the solution to $\dot{\tilde{p}}_t = f(\tilde{p}_t)$ with initial condition $\tilde{p}_0 = p_0$. It is easy to show that $p_t \geq \tilde{p}_t$ at all times, and that $\tilde{p}_t$ converges monotonically to a steady state $\tilde{p}^* \in (0,1)$. The condition $\mu < \left(\frac{\lambda_0+\lambda_1+r}{\lambda_1+r}\right) r$ ensures that this steady state is above $\frac{\lambda_0}{\lambda_0+\lambda_1+r}$. Therefore, p_t does not cross this threshold.

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